

The paintings at El Castillo in Spain, dating from around 41,000YA, are among the oldest known cave art. Made using natural pigments, the paintings include depictions of horses and bison, although the very earliest are abstract disks and dots.

THE FIRST SIGNIFICANT

SCIENTIFIC ADVANCE was the production of stone tools. Around 2.5 million years ago (MYA), early hominids (either *Homo habilis* or *Australopithecus*) began to modify cobbles by striking them with another stone, thus removing flakes of stone and creating a sharp edge—a method known as **hard-hammer percussion**. These early pebble tools, or choppers, are known as **Oldowan tools**. They were used for dismembering killed animals, cracking bones for the marrow, and scraping hides. Oldowan technology spread throughout Africa, where it lasted until around 1.7 MYA.

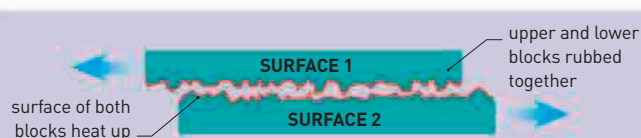
Early hominids must have seen and understood the power of fire by observing wildfires caused by lightning strikes. They may have



sharp edge where stone flake struck off

Oldowan tool

Choppers like this were the earliest stone tools. They were suitable for tasks such as cutting animal hides.



GENERATING HEAT FROM FRICTION

Rubbing two surfaces together causes the kinetic movement energy of the rubbing motion to be transferred to the atoms in the surfaces. This process, known as friction, causes the atoms to heat up. The smoother the surfaces, the more heat is generated; in extreme cases this can cause nearby material to catch fire.

lit branches from these fires to use as weapons against predators or to provide light and heat. There is possible **evidence for sporadic controlled use of fire** from around 1 MYA, with evidence of regular use from around 400,000 YA. Finds at Gesher Benot Ya'aqov in Israel (790,000 YA) show signs of the **active use of fire**.

Early humans were able to use devices such as fire plows or fire drills to produce their own fire with friction. Fire was important for warmth and protection, splitting stones, hardening the points of wooden tools, and cooking.

Heating food breaks down proteins, which makes it easier to digest. It also protects food from

decay and extends the range of edible resources to include plants containing toxins that can be broken down by heat. The earliest evidence of cooking comes from sites such as Gesher Benot Ya'aqov in Israel (790,000 YA), where concentrations of burnt seeds and wood were found.



Making fire

Early humans probably made fire using a fire drill or fire plow, which generates heat by rubbing two pieces of wood together. The heat causes wood dust to ignite and this can then be used to light larger kindling.

flame generated in kindling such as twigs

Around 1.76 MYA, more advanced stone tools began to appear. Unlike Oldowan tools, **Acheulian tools**, particularly the multipurpose handaxe, were deliberately shaped. **Hard-hammer percussion** (striking off flakes with a hammerstone) was used to rough out the tool's shape. It was then refined by removing smaller flakes using a soft hammer of bone or antler.

Mousterian tools are particularly associated with Neanderthals and occurred from c.300,000 YA. They include sharp-edged Levallois flakes that were struck off a prepared

Upper Paleolithic leaf point

This skillfully crafted tool was made by flaking small pieces off a larger core using a sharp piece of bone or antler to apply pressure.



core (see panel, opposite), and a wide range of flake tools—such as knives, spear points, and scrapers—shaped for different purposes.

In the late Middle and Upper Paleolithic (c.35,000–10,000 YA), a new technique, indirect percussion, allowed for many blades to be struck from a single core. The final stage of tool development first appeared c.70,000 YA and became widespread post-glacially from about 10,000 YA. It involved **microliths**—tiny flakes and blades for use in composite tools.

The earliest weapons were rocks or handaxes, but by about 400,000 BCE early people had adapted **sticks for use as spears**. At first, these had sharpened wooden ends, but by around 200,000 BCE stone points started being attached to create more effective weapons. The bow was probably first developed around 64,000 BCE, but the earliest examples found

c.2.5 MYA Earliest Oldowan stone tools are made

c.1.76 MYA Earliest Acheulian handaxes

c.790,000 YA First controlled use of fire, found at Benot Ya'aqov, Israel

c.500,000 BCE Earliest antler tools, found at Boxgrove, England

c.400,000 BCE Oldest wooden spears, found at Gönningen, Germany

125,000 BCE Mousterian flake tools made using Levallois technique become predominant in Europe

c.64,000 BCE Bow is first developed

c.39,000 BCE Earliest known cave paintings, at El Castillo, Spain

c.30,000 BCE Earliest bone needles, found in Europe

c.30,000 BCE Dogs first domesticated